

SYDNEY, NS, Canada

WMO: 717070

Lat: 46.163N

Lon: 60.043W

Elev: 62

StdP: 100.58

Time Zone: -4.00 (NAA)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-16.6	-14.2	-21.8	0.5	-15.5	-19.2	0.7	-13.5	15.8	-3.2	14.1	-4.5	4.2	250	0.692

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.9	27.7	20.5	26.0	19.6	24.4	18.8	21.8	25.7	20.8	24.3	19.9	22.9	5.6	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.5	15.3	23.7	19.6	14.4	22.6	18.7	13.6	21.6	64.0	25.8	60.5	24.5	57.3	22.9	25.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation	n=5 years		n=10 years		n=20 years		n=50 years		DB	WB
Min	Max			Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
(4) 12.9	11.3	10.0		-19.4	31.1	3.2	1.3	-21.7	32.0	-23.6	32.8	-25.4	33.6	-27.7	34.5	(4)
(5)				-19.9	23.6	3.1	0.9	-22.2	24.2	-24.0	24.8	-25.7	25.3	-28.0	25.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.4	-4.5	-5.1	-2.1	2.7	7.9	13.2	18.4	18.6	14.7	9.1	4.2	-0.7		
	DBStd	9.07	4.94	4.92	4.34	3.52	3.71	3.85	3.03	2.76	3.55	3.55	3.99	4.17		
	HDD10.0	2141	449	422	374	221	88	12	0	0	4	61	179	331		
	HDD18.3	4432	707	655	632	469	322	159	37	32	119	287	423	589		
	CDD10.0	842	0	0	0	2	24	108	261	266	144	33	5	0		
	CDD18.3	93	0	0	0	0	0	6	39	39	9	0	0	0		
	CDH23.3	571	0	0	0	0	6	61	258	201	46	0	0	0		
	CDH26.7	94	0	0	0	0	1	8	44	36	6	0	0	0		
	Wind	WSAvg	4.9	5.5	5.5	5.5	5.2	4.6	4.4	4.1	4.0	4.4	4.7	5.1	5.5	(14)
(15) Precipitation	PrecAvg	1523	156	134	143	123	100	92	91	97	111	142	163	176		
	PrecMax	2050	259	228	254	278	243	186	222	225	334	279	351	315		
	PrecMin	1170	60	55	55	25	23	27	18	21	43	39	74	66		
	PrecStd	176	43	49	45	53	50	43	52	43	52	55	69	60		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.2	9.4	11.6	17.4	23.5	27.7	30.1	30.1	27.2	21.2	16.5	12.6		
		MCWB	10.5	8.6	7.7	12.0	15.4	19.0	21.9	21.9	20.5	16.9	15.1	11.8		
	2%	DB	7.8	6.4	8.1	13.0	20.6	25.1	27.8	27.4	24.4	18.8	14.6	10.2		
		MCWB	7.4	5.5	6.7	9.7	13.9	18.2	21.0	20.4	19.2	16.0	13.5	9.6		
	5%	DB	4.9	3.8	5.9	10.7	17.5	22.7	26.2	25.5	22.1	16.9	13.0	7.6		
		MCWB	4.3	2.8	4.2	7.9	12.1	16.5	19.8	19.6	18.2	14.6	12.0	6.9		
	10%	DB	2.3	1.6	3.8	8.6	14.7	20.4	24.5	23.8	20.4	15.0	10.8	5.2		
		MCWB	1.5	0.7	2.3	6.3	10.5	15.6	19.0	18.8	17.4	13.0	9.6	4.6		
	0.4%	WB	10.6	8.8	9.1	13.6	17.7	21.1	23.0	23.1	21.8	18.2	15.5	12.1		
		MCDB	11.1	9.1	10.1	16.0	21.6	26.0	28.0	27.4	24.9	19.7	16.2	12.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	7.5	5.8	6.9	10.5	15.0	19.3	22.0	21.7	20.5	16.7	13.7	9.7		
		MCDB	7.9	6.4	7.7	12.1	18.9	23.1	26.5	25.1	23.2	18.2	14.3	10.3		
	5%	WB	4.4	3.0	4.5	8.5	13.0	17.7	20.9	20.7	19.3	15.0	12.1	7.0		
		MCDB	4.9	3.8	5.7	10.1	16.2	21.0	24.7	24.0	21.3	16.5	13.0	7.6		
	10%	WB	1.6	0.7	2.4	6.4	11.1	16.1	19.8	19.8	18.0	13.2	9.9	4.7		
		MCDB	2.2	1.4	3.5	8.3	14.2	19.9	22.9	22.7	19.9	14.8	10.6	5.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.6	7.8	7.3	7.6	9.3	9.6	9.5	8.9	8.8	7.5	6.5	6.2		
		MCDBR	10.8	10.3	10.3	12.1	14.6	13.7	12.1	11.4	11.0	10.0	9.2	9.7		
		MCWBR	11.1	10.0	8.8	9.0	9.1	8.0	6.2	6.0	6.8	7.6	8.8	9.9		
	5% WB	MCDBR	10.8	10.4	10.0	10.9	12.9	11.6	10.3	9.8	9.0	9.0	8.9	9.5		
		MCWBR	11.2	10.4	9.3	9.3	9.1	7.6	5.8	5.8	6.6	7.8	9.0	9.9		
(40) Clear-Sky Solar Irradiance	taub		0.303	0.297	0.329	0.359	0.370	0.390	0.420	0.394	0.364	0.346	0.324	0.300		
	taud		2.368	2.401	2.366	2.391	2.430	2.396	2.350	2.428	2.488	2.500	2.490	2.443		
	Ebn at Noon		791	884	898	899	897	877	843	854	853	817	768	759		
	Edh at Noon		74	86	104	111	111	116	120	107	92	79	66	62		
(44) All-Sky Solar Radiation	RadAvg		1.04	1.85	3.16	4.23	5.10	5.40	5.36	4.83	3.78	2.35	1.24	0.80		
	RadStd		0.08	0.20	0.28	0.33	0.46	0.49	0.36	0.34	0.28	0.20	0.15	0.07		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	+0.41	+0.61	N/A	N/A	N/A	N/A
(47) Regional (16 neighbors)		N/A	N/A	N/A	N/A	N/A	+0.34	N/A	N/A	N/A	N/A

Nomenclature: See separate page